1.2 - Give the formal description of the machines M_1 and M_2

For M_1 : $(\{q_1, q_2, q_3\}, \{a, b\}, \delta_1, q_1, \{q_2\})$

$\delta_1(q_1, b) = q_1$ $\quad$ $\delta_1(q_1, a) = q_2$

$\delta_1(q_2, a) = q_3$ $\quad$ $\delta_1(q_2, b) = q_3$

$\delta_1(q_3, a) = q_2$ $\quad$ $\delta_1(q_3, b) = q_1$

For M_2 : $(\{q_1, q_2, q_3, q_4\}, \{a, b\}, \delta_2, q_1, \{q_1, q_4\})$

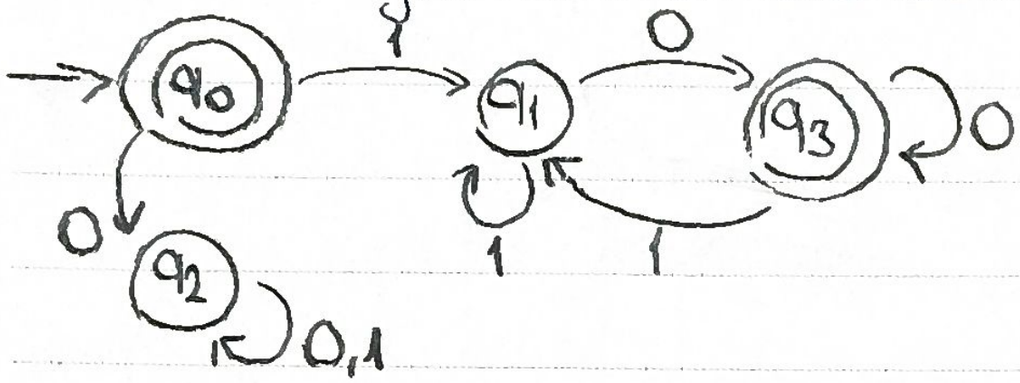
	q_1	q_2	q_3	q_4
q_1	a	b	$\emptyset$	$\emptyset$
q_2	$\emptyset$	$\emptyset$	a	b
q_3	b	a	$\emptyset$	$\emptyset$
q_4	$\emptyset$	$\emptyset$	a	b

$= \delta_2$

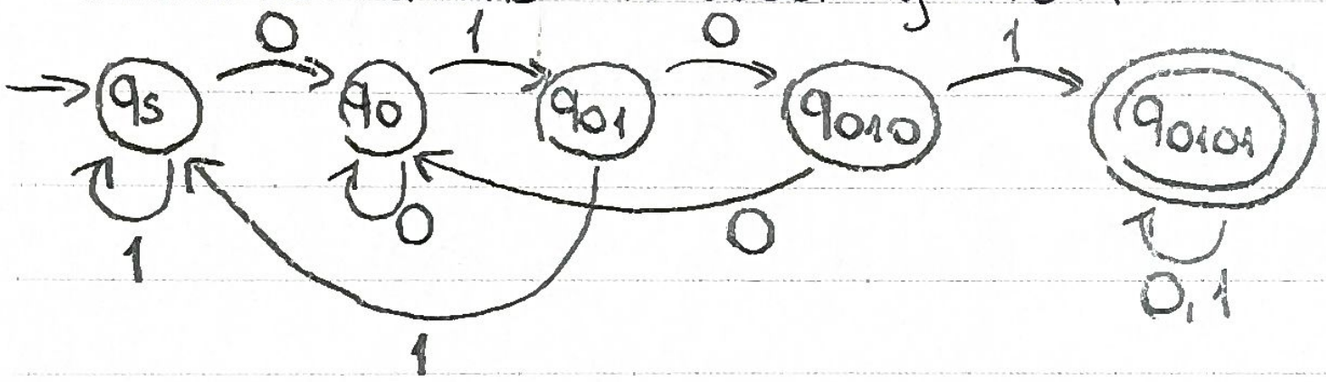
$(Q, \Sigma, \delta, s, F)$ 5-tuple

1.6 - Give state diagrams of DFAs recognizing the following languages. $\Sigma = \{0,1\}$

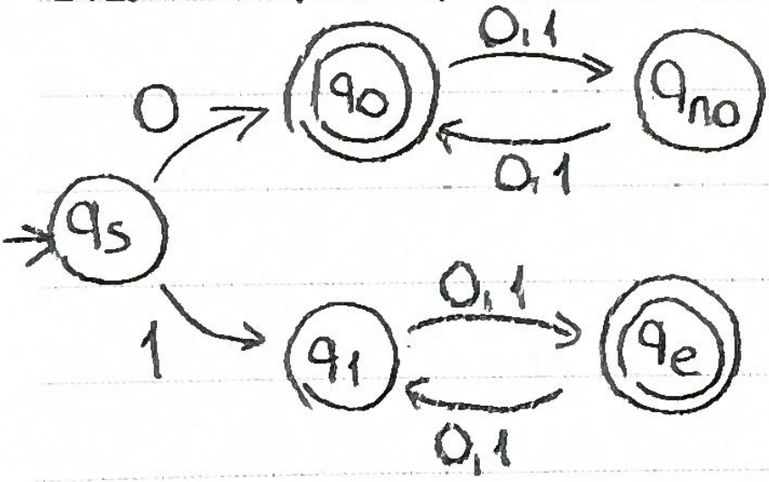
a. $\{w \mid w$ begins with a 1 and ends with a 0$\}$



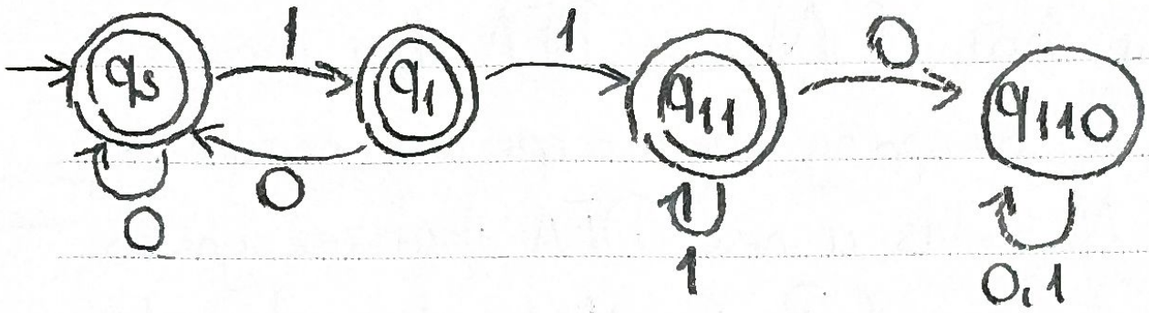
c. $\{w \mid w$ contains the substring 0101$\}$



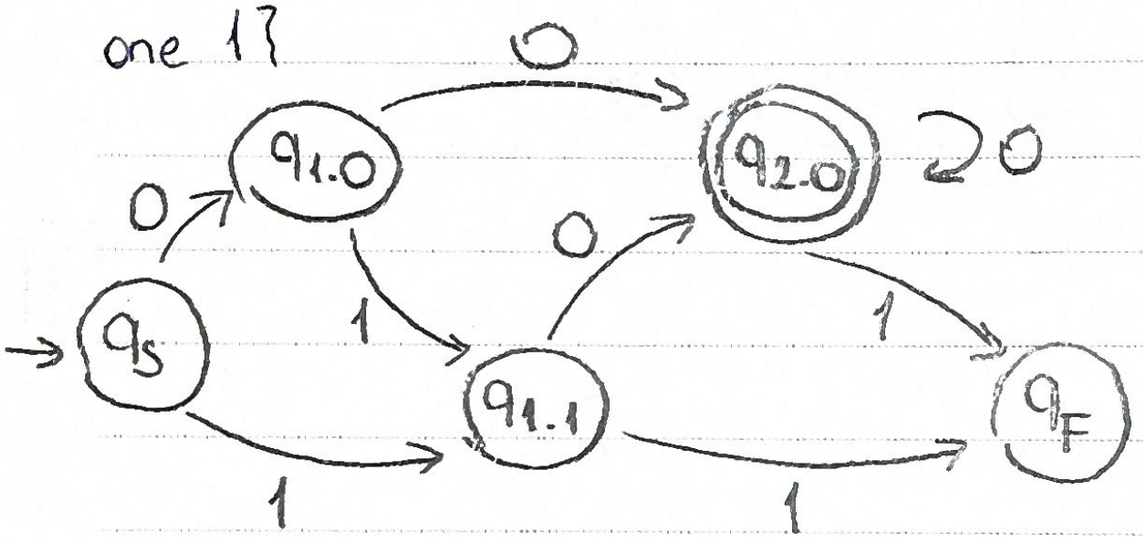
e. $\{w \mid w$ starts with 0 and has odd length, or starts with 1 and has even length$\}$



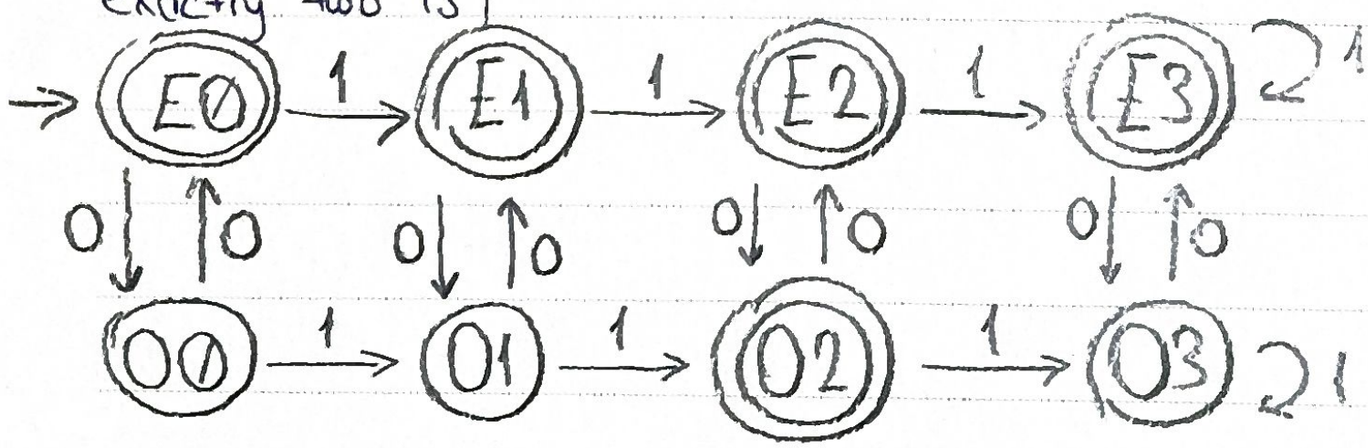
f. {w | w doesn't contain the substring 110}



j. {w | w contains at least two 0s and at most one 1}



l. {w | w contains an even number of 0s, or contains exactly two 1s}



E = even
O = odd

1.14

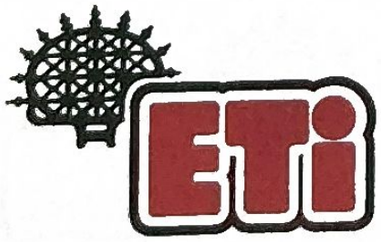
a) Show that, if M is a DFA that recognizes language B, swapping the accept and nonaccept states in M yields a new DFA that recognizes the complement of B. Conclude that the class of regular languages is closed under compliment.

$M = (Q, \Sigma, \delta, q_0, F)$

$M' = (\ ,\ ,\ ,\ ,\ ,\ Q - F)$

$L(M') = \overline{L(M)}$

a DFA recognizes ↙ ↘ a new DFA recognizes



1.16 - Use the construction given in Theorem 1.39 to convert the following two NFA to DFA.

a) $Q_D = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$

$s_D = \{1\}$ $\quad F_D = \{\{1\}, \{1,2\}\}$

$\delta(\{1\}, a) = \{1,2\}$ $\quad \delta(\{1\}, b) = \{2\}$

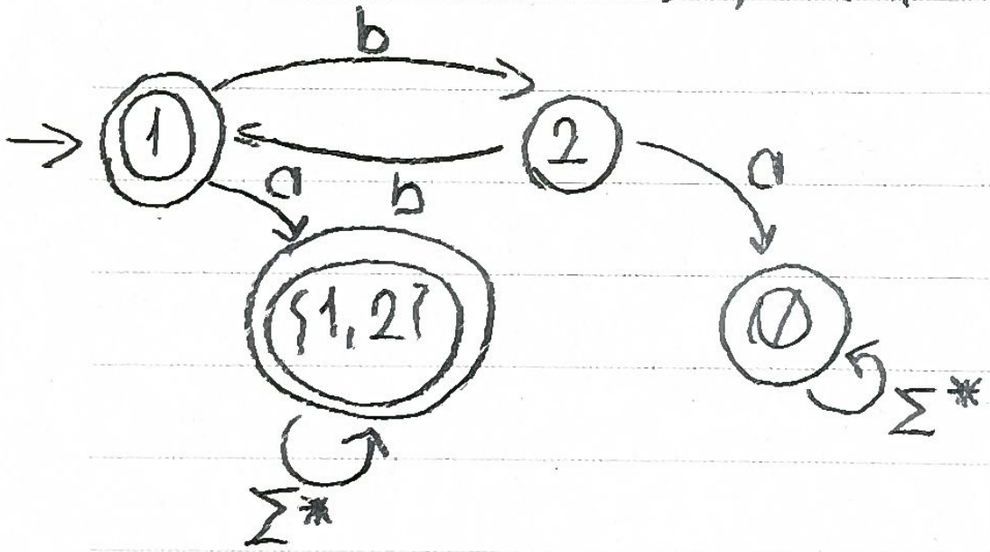
$\delta(\{2\}, a) = \emptyset$ $\quad \delta(\{2\}, b) = \{1\}$

$\delta(\{1,2\}, a) = \delta(\{1\}, a) \cup \delta(\{2\}, a)$

$= \{1,2\} \cup \emptyset = \{1,2\}$

$\delta(\{1,2\}, b) = \delta(\{1\}, b) \cup \delta(\{2\}, b)$

$= \{2\} \cup \{1\} = \{1,2\}$



accept = $\{\{1\}, \{1,2\}\}$ $\quad G_1$

nonaccept = $\{\{2\}, \emptyset\}$ $\quad G_2$

	a	b
$\emptyset$	G_2	G_2
$\{1\}$	G_1	G_2
$\{2\}$	G_2	G_1
$\{1,2\}$	G_1	G_1

no pair

no DFA minimalization

b) $\Theta_D = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{1,2,3\}\}$

$s = \{1\}$; $\{1\} \xrightarrow{\varepsilon} \{2\}$, $s_D = \{1\} \cup \{2\} = \{1,2\}$

$F = \{2\}$ all states include $\{2\}$ $F_D = \{\{2\}, \{1,2\}, \{2,3\}, \{1,2,3\}\}$

$\delta(\{1\}, a) = \{3\}$ $\delta(\{1\}, b) = \emptyset$

$\delta(\{2\}, a) = \{1,2\}$ $\delta(\{2\}, b) = \emptyset$

$\delta(\{3\}, a) = \{2\}$ $\delta(\{3\}, b) = \{2,3\}$

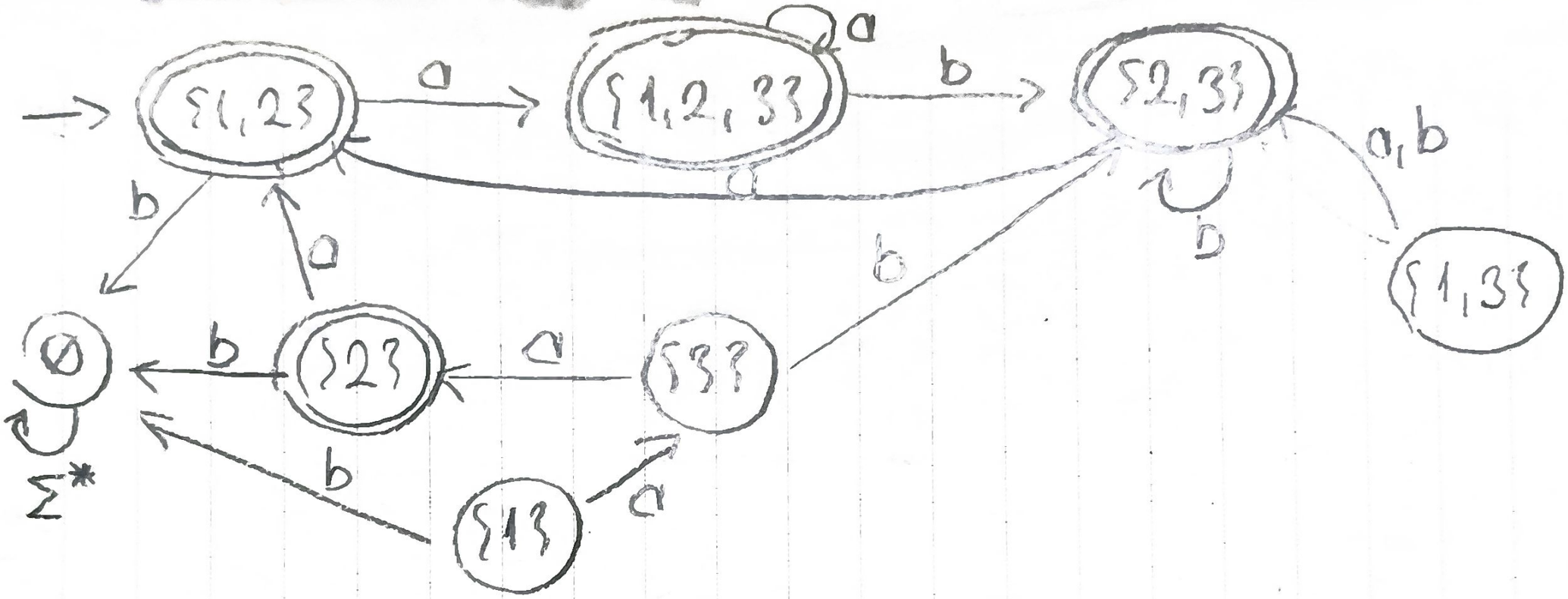
$\delta(\{1,2\}, a) = \{1,2,3\}$ $\delta(\{1,2\}, b) = \emptyset$

$\delta(\{2,3\}, a) = \{1,2\}$ $\delta(\{2,3\}, b) = \{2,3\}$

$\delta(\{1,2,3\}, a) = \{1,2,3\}$

$\delta(\{1,2,3\}, b) = \{2,3\}$

$\delta(\{1,3\}, a) = \{2,3\}$ $\delta(\{1,3\}, b) = \{2,3\}$



The states reached from start state = {1,2}, ∅, {1,2,3}, {2,3}

others = {2}, {3}, {1}, {1,3}

remove them for the DFA minimization

$A^{ccept} = \{1,2\}, \{2,3\}, \{1,2,3\}$ $\quad \sigma_1$ $\quad \sigma_2$

$N = \emptyset$

	a	b
$\{1,2\}$	σ_1	σ_2
$\{2,3\}$	σ_1	σ_1
$\{1,2,3\}$	σ_1	σ_1

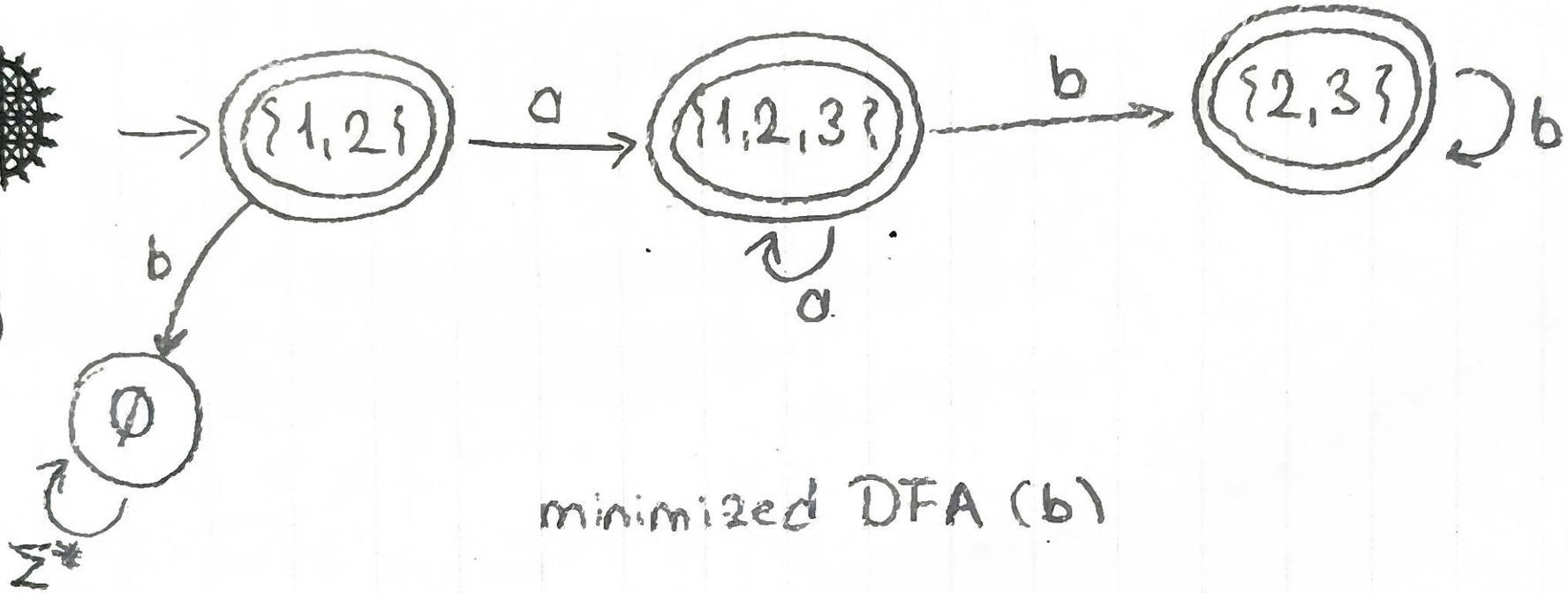
$\emptyset$ | $\{1,2\}$ | $\{2,3\}, \{1,2,3\}$

σ_1 $\quad \sigma_2$ $\quad \sigma_3$

	a	b
$\emptyset$	σ_1	σ_1
$\{1,2\}$	σ_3	σ_1
$\{2,3\}$	σ_2	σ_3
$\{1,2,3\}$	σ_3	σ_3

] their rotation did not match

$\rightarrow \emptyset$ | $\{1,2\}$ | $\{2,3\}$ | $\{1,2,3\}$



minimized DFA (b)

1.28- Convert the following regular expressions to NFAs using the procedure given in Theorem 1.54. In all parts $\Sigma = \{a, b\}$.

a) $a(abb)^* \cup b$

a



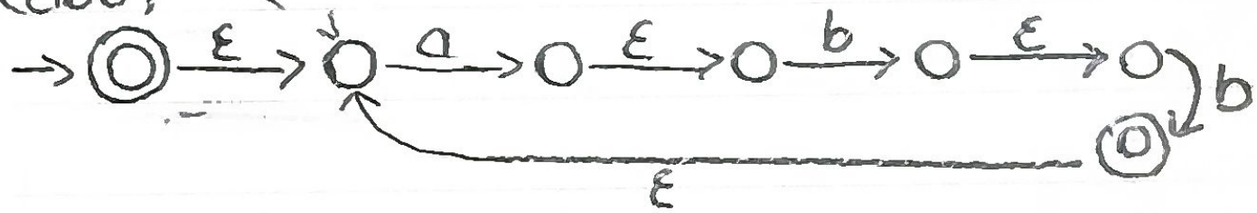
b



abb



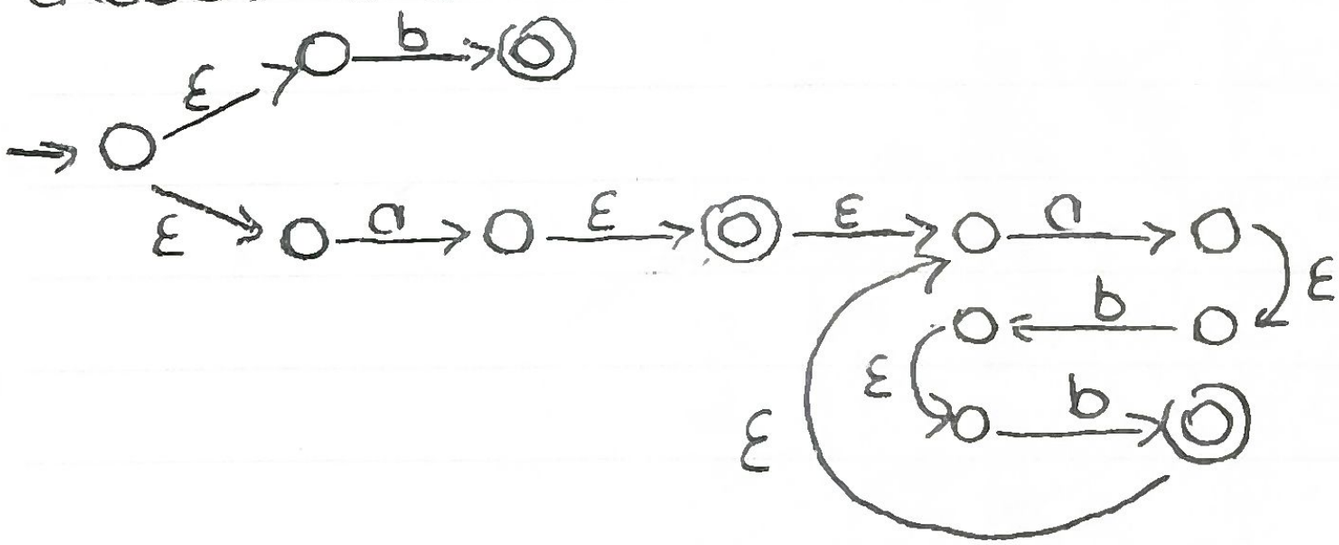
$(abb)^*$



$a(abb)^*$



$a(abb)^* \cup b$

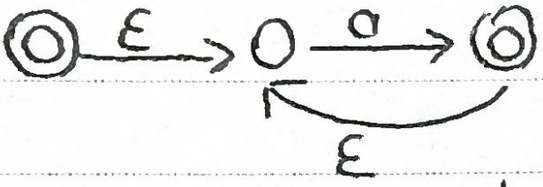


b) $a^* \cup (ab)^*$

a



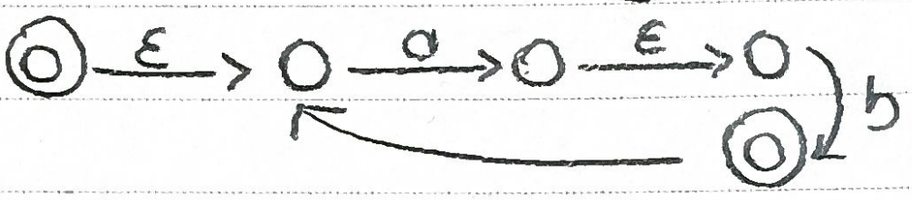
a^*



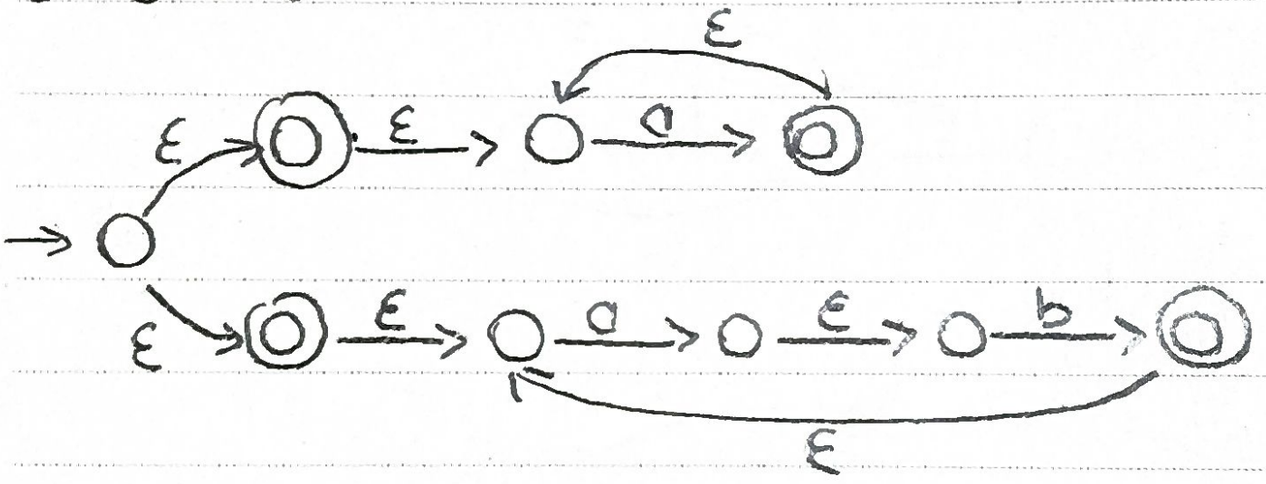
ab



$(ab)^*$



$a^* \cup (ab)^*$

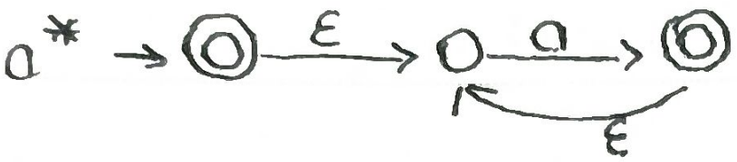


c) $(a \cup b^*)\, a^*\, b^*$

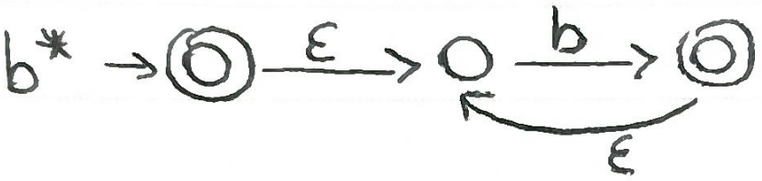
a



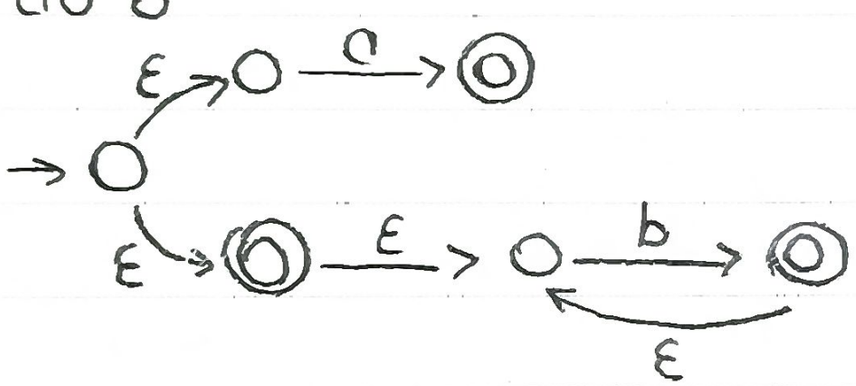
a^*



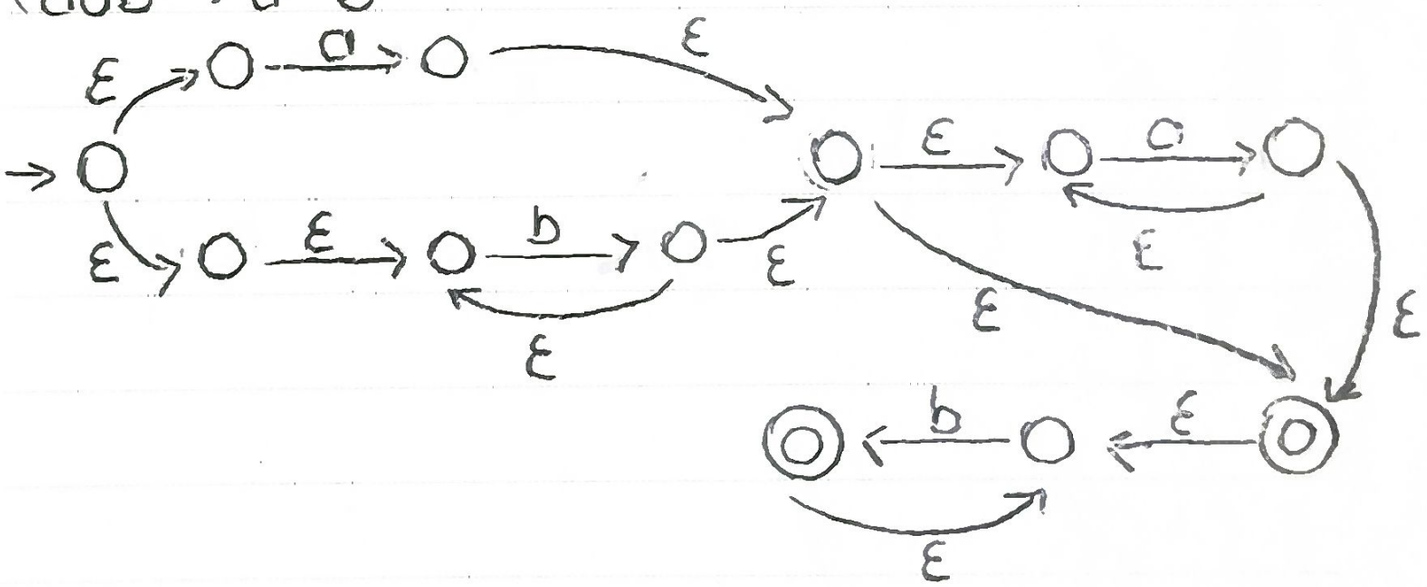
b^*



$a \cup b^*$



$(a \cup b^*)\, a^*\, b^*$



1.29. Use the pumping lemma to show that the following languages are not regular.

a) $A_1 = \{0^n 1^n 2^n \mid n \geq 0\}$

$s = 0^p 1^p 2^p$

$0^p \mid 1^p \mid 2^p$ ② ✗

$0^p \mid 1^p 2^p$ ② ③ ✗

$(0^p)^i 1^p 2^p$

$i=0 \quad = 1^p 2^p \notin A_1$ contradiction

$i=1 \quad = 0^p 1^p 2^p \in A_1$

$i=2 \quad = 0^p 0^p 1^p 2^p \notin A_1$ contradiction

As there is a contradiction, A_1 is not regular.

① $xy^i z \in A_1$

② $|y| > 0$

③ $|xy| \leq p$

b) $A_2 = \{www \mid w \in \{a,b\}^*\}$

$s = a^p a^p a^p$

$a^p \mid a^p \mid a^p$ ① ✗

$a^p \mid a^p a^p$ ② ③ ✗

$(a^p)^i a^p a^p$

$i=0 \quad = a^p a^p \notin A_2$ contradiction

$i=1 \quad = a^p a^p a^p \in A_2$

$i=2 \quad = a^p a^p a^p a^p \notin A_2$ contradiction

As there is a contradiction, A_2 is not regular.

① $xy^i z \in A_1$

② $|y| > 0$

③ $|xy| \leq p$

c) $A_3 = \{a^{2^n} \mid n \geq 0\}$

$a^{2^0} = 1 \quad a^1 = a$

$a^{2^1} = 2 = a^2 = aa$

$a^{2^2} = 4 = a^4$

$a^{2^3} = 8 = a^8$

$a^{2^4} = 16 = a^{16}$

$a^{2^{(p+1)}} = a^{2^p} \cdot a^2$

① $xy^i z \in A_3$

② $|y| > 0$

③ $|xy| \leq p$

?

1.30 - Describe the error in the following "proof" that 0^*1^* is not a regular language. The proof is by contradiction.

Assume that 0^*1^* is regular. Let p be the pumping length for 0^*1^* given by the pumping lemma. Choose s to be the string 0^p1^p. ~~You know~~

$s = 0^p1^p$

0^p | 1^p ③ ②

$(0^p)^i 1^p$

① $xy^iz \in A$

② $|y| > 0$

③ $|xy| \leq p$

$i=0$ $= 1^p \in A$ (* includes ε as well)

$i=1$ $= 0^p1^p \in A$

$i=2$ $= 0^{2p}1^p \in A$

In example 1.73 the class $B = \{0^n1^n \mid n \geq 0\}$. That's why $i=0,2$ outputs does not in B. However, our class is 0^*1^*. In this class, because of *, all of the outputs included by 0^*1^*.